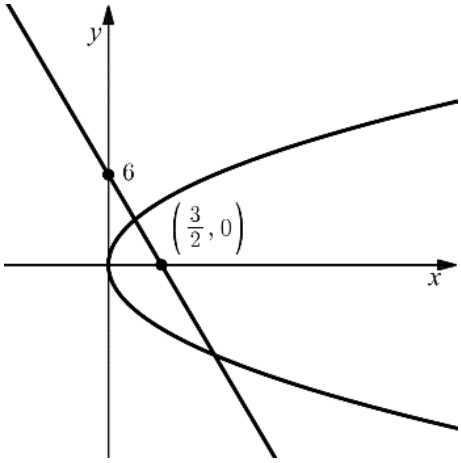
1) a)



1) b)

$(6-4x)^2 = 12x$

$36 - 48x + 16x^2 = 12x$

$16x^2 - 60x + 36 = 0$

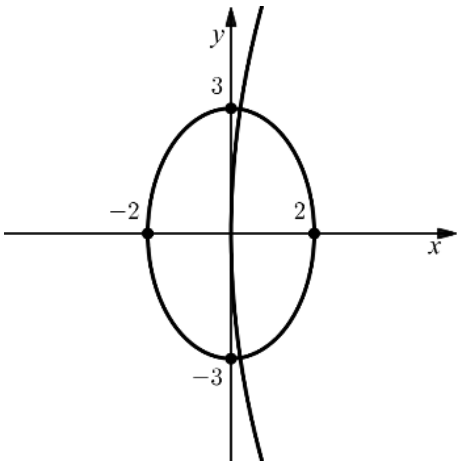
$\therefore x = \frac{3}{4}$ or $x = 3$

So $\left(\frac{3}{4}, 3\right)$ and $(3, -6)$

1) c)

$\sqrt{9^2 + \left(\frac{9}{4}\right)^2} = \frac{9\sqrt{17}}{4}$

2) a)



2) b)

$\frac{x^2}{4} + \frac{40x}{9} = 1$

$9x^2 + 160x = 36$

$9x^2 + 160x - 36 = 0$

$(9x - 2)(x + 18) = 0$

$x = \frac{2}{9}$ or $x = -18$

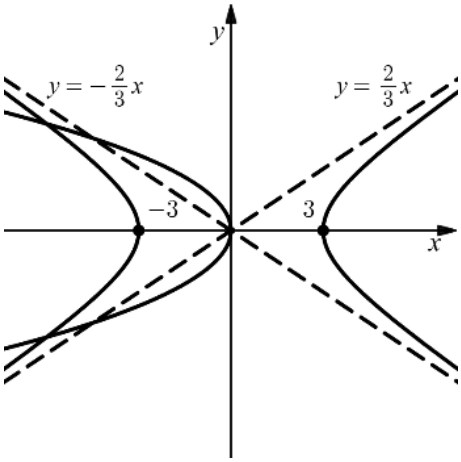
So since x is positive

$x = \frac{2}{9}$

$y^2 = 40 \times \frac{2}{9} \Rightarrow y = \pm\frac{4\sqrt{5}}{3}$

$\left(\frac{2}{9}, -\frac{4\sqrt{5}}{3}\right)$ and $\left(\frac{2}{9}, \frac{4\sqrt{5}}{3}\right)$

3) a)



3) b)

$\frac{x^2}{9} - \frac{(-2x)}{4} = 1$

$4x^2 + 18x = 36$

$4x^2 + 18x - 36 = 0$

$2x^2 + 9x - 18 = 0$

$(x + 6)(2x - 3) = 0$

$x = -6$ or $x = \frac{3}{2}$

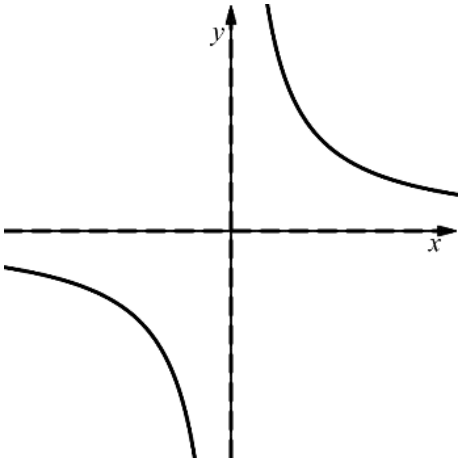
So since x is negative

$x = -6$

$y^2 = -2 \times -6 \Rightarrow y = \pm 2\sqrt{3}$

$(-6, -2\sqrt{3})$ and $(-6, 2\sqrt{3})$

4) a)



4) b)

$x\left(-kx+8\sqrt{2}\right)=16$

$-kx^2+8\sqrt{2}x-16=0$

$kx^2-8\sqrt{2}x+16=0$

$\therefore x=\frac{8\sqrt{2}\pm\sqrt{128-64k}}{2k}=\frac{4\sqrt{2}-4\sqrt{2-k}}{k}$

4) c)

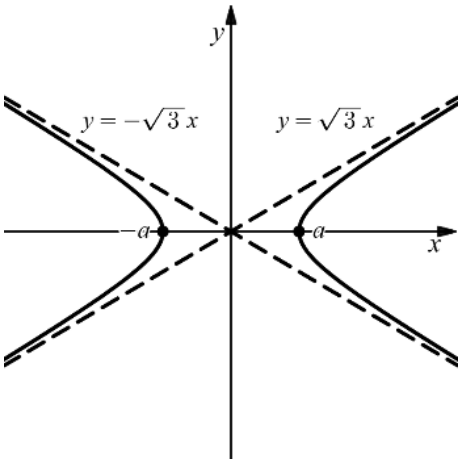
$2-k=0$ so $k=2$

5) a)

$y=\pm\frac{\sqrt{3a^2}}{a}x=\pm\sqrt{3}a$

$y=\sqrt{3}a$ and $y=-\sqrt{3}a$

5) b)



5) c)

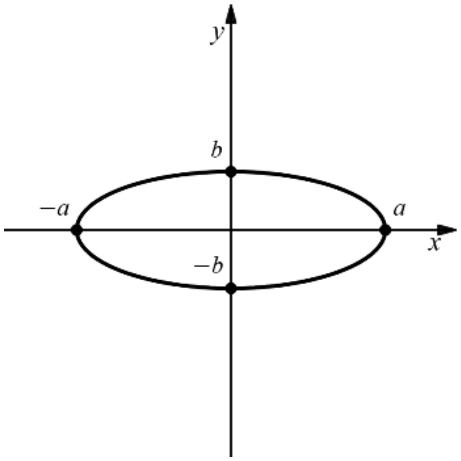
$\frac{y^2}{3a^2} = \frac{x^2}{a^2} - 1 \Rightarrow y^2 = 3x^2 - 3a^2$

$\pi \int_a^{2a} 3x^2 - 3a^2 \, \mathrm{d}x = \pi[x^3 - 3a^2x]_a^{2a}$

$= \pi[(8a^3 - 6a^3) - (a^3 - 3a^3)]$

$= 4a^3\pi$ units^3

6) a)



6) b)

$\frac{x^2}{a^2} = 1 - \frac{y^2}{b^2} \Rightarrow x^2 = a^2 - \frac{a^2}{b^2}y^2$

$\pi \int_0^b a^2 - \frac{a^2}{b^2}y^2 \, \mathrm{d}y = \pi \left[a^2y - \frac{a^2}{3b^2}y^3\right]_0^b$

$= \pi \left[\left(a^2b - \frac{1}{3}a^2b\right) - (0)\right]$

$= \frac{2}{3}a^2b\pi$ units^3

7)

$m^2x^2 = 12x - 12k$

$m^2x^2 - 12x + 12k = 0$

For it to be a tangent, $b^2 - 4ac = 0$

$(-12)^2 - 4m^2 \times 12 = 0$

$144 - 48m^2k = 0$

$48m^2k = 144$

$m^2 = \frac{3}{k}$

$\therefore m = \pm\sqrt{\frac{3}{k}}$ as req.

8)

$\frac{x^2}{4} + \frac{(mx+5)^2}{16} = 1$

$4x^2 + (mx+5)^2 = 16$

$4x^2 + m^2x^2 + 10mx + 25 = 16$

$(4+m^2)x^2 + 10mx + 9 = 0$

For it to be a tangent, $b^2 - 4ac = 0$

$(10m)^2 - 4(4+m^2)(9) = 0$

$100m^2 - 144 - 36m^2 = 0$

$64m^2 - 144 = 0$

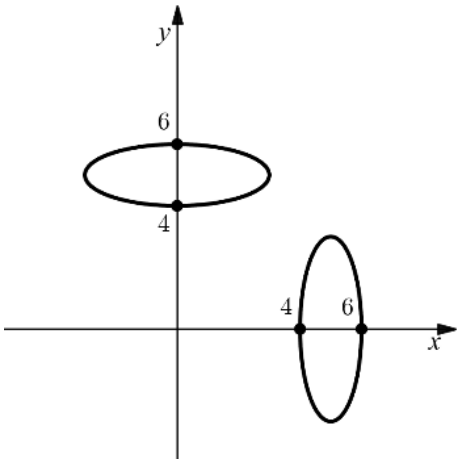
$m^2 = \frac{9}{4}$

$m = -\frac{3}{2}$ or $m = \frac{3}{2}$

9) a)

E_1: when $y = 0$, $(x-5)^2 = 1$ so $x = 4$ or $x = 6$

E_2: when $x = 0$, $(y-5)^2 = 1$ so $y = 4$ or $y = 6$



9) b)

Tangent line for E_1:

$(x-5)^2+\frac{(k-x)^2}{9}=1$

$9(x-5)^2+(k-x)^2=9$

$9x^2-90x+225+k^2-2kx+x^2=9$

$10x^2-(90+2k)x+(216+k^2)=0$

For it to be a tangent, $b^2-4ac=0$

$\left(-(90+2k)\right)^2-4\times10(216+k^2)=0$

$4k^2+360k+8100-40k^2-8640=0$

$-36k^2+360k-540=0$

$k^2-10k+15=0$

$k=5+\sqrt{10}$ or $k=5-\sqrt{10}$

9) c)

Tangent line for E_2:

$\frac{(k-y)^2}{9}+(y-5)^2=1$

Which is the same as seen in part b) to give $10y^2-(90+2k)y+(216+k^2)=0$

$\therefore k=5+\sqrt{10}$ or $k=5-\sqrt{10}$

10)

Given the general point $P(x,y)$

Distance to point: $\sqrt{x^2+(y+4)^2}$

Distance to line: $|y-2|$

$\sqrt{x^2+(y+4)^2}=|y-2|$

$x^2+(y+4)^2=(y-2)^2$

$x^2+y^2+8y+16=y^2-4y+4$

$x^2=-12y-12$

$x^2=-12(y+1)$

11)

Given the general point $P(x, y)$

Distance to point: $\sqrt{x^2 + (y-9)^2}$

Distance to line: $|x - 1|$

$\sqrt{(x-9)^2 + y^2} = 3|x-1|$

$(x-9)^2 + y^2 = 9(x-1)^2$

$x^2 - 18x + 81 + y^2 = 9x^2 - 18x + 9$

$y^2 = 8x^2 - 72$

$8x^2 - y^2 = 72$

$\frac{x^2}{9} - \frac{y^2}{72} = 1$